

The Electronic Structure of Conjugated Systems. VI

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If g_1 is large compared with g ,

$$\frac{\eta}{\eta_0} = \cosh \sqrt{\left\{ \frac{4\pi s}{\lambda} \frac{(\rho_2 - \rho_1)}{(\rho_2 + \rho_1)} \right\}} = \cosh \sqrt{\left\{ 4\pi m \frac{(\rho_2 - \rho_1)}{(\rho_2 + \rho_1)} \right\}}, \quad (32)$$

where m is the number of wave-lengths that the liquid has descended. Thus, when $g_1 \gg g$, the amplification of an unstable disturbance depends only on the ratio of the densities of the two fluids and the number of wave-lengths through which they have descended.

The experiments described in part II were designed to test the above theoretical conclusions and to find out how the unstable surface behaves when the amplitude of the disturbance becomes so great that the analysis based on the assumption that it is small ceases to be applicable.

REFERENCE

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The electronic structure of conjugated systems. VI

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This part reconsiders the previously neglected non-orthogonality of atomic orbitals. New generalized definitions of fractional bond order and charge distribution are introduced, and it is shown that under certain conditions, which are satisfied by most conjugated hydrocarbons, the overlap factor is completely irrelevant in calculating bond orders and charges. For heteromolecules this is not the case, and substantial changes may sometimes be made by its inclusion. The various polarizability coefficients introduced in earlier papers are always affected, but numerical results suggest that these changes are not very large.

INTRODUCTION

Previous papers in this series (Coulson & Longuet-Higgins 1947–8, referred to as I to V) have discussed in quite general terms the bond orders and charge distribution in conjugated systems; and have introduced various polarizability coefficients whose values showed how a molecule was likely to be affected by a change in the environment of any part of it. Applications were made to the law of alternating polarity in chemical reactions, to the vibrations of condensed molecules such as benzene and naphthalene, to the conjugating power of unsaturated free radicals, and to the differences between hydrocarbon systems such as naphthalene and heterosystems such as quinoline, in which a CH group has been replaced by a N atom. But in all this work it was explicitly supposed that all overlap integrals $S_{\mu\nu} = \int \phi_\mu \phi_\nu d\tau$ between distinct atomic orbitals ϕ_μ and ϕ_ν were identically zero. Such an approximation has almost universally been made hitherto, despite the fact that numerical calculations

show that for two adjacent carbon π -electron orbitals $S_{\mu\nu}$ is in the region of 0.25. This is very far from being negligible in comparison with the $S_{\mu\mu}$, which are all unity, since each ϕ_μ is normalized. The only general discussions of the possibility of including $S_{\mu\nu}$ of which we are aware are for hydrocarbons by Wheland (1941), who showed how to correct the energy values previously obtained by neglect of $S_{\mu\nu}$, and for graphitic crystallites by Bradburn, Coulson & Rushbrooke (1948), who were also chiefly concerned with energy values. Both of these papers dealt with only one kind of unsaturated atom, here carbon. It is our purpose in this paper to discuss quite generally the significance of $S_{\mu\nu}$ for molecules in which the unsaturated atoms are not all of one kind. This will involve us in new definitions of bond order and charge density. Under certain circumstances, of which the contents of the papers referred to above are particular examples, it will be shown that inclusion of $S_{\mu\nu}$ makes no difference whatever to charge density or bond orders, though it does always make numerical changes in the polarizability coefficients. However, these latter changes appear often to be small. But with heteromolecules, both charges and bond orders are sensitive to the values of the $S_{\mu\nu}$.

NOTATION

So far as possible we shall use the notation of previous papers of this series. But certain changes are necessitated by our extension of earlier theorems. In particular, we shall denote different atoms by a Greek suffix $\mu, \nu, \rho, \dots$ and a molecular orbital (m.o.) by a Latin suffix $i, j, \dots$. In the usual LCAO approximation (see earlier papers in this series), we expand each m.o. ψ_j in the form

$$\psi_j = \sum_{\mu=1}^n x_{j\mu} \phi_\mu. \quad (1)$$

When there is no doubt about which m.o. we are discussing, we can drop the suffix j and write simply x_μ instead of $x_{j\mu}$. There are n atoms in the molecule, and the coefficients $x_{j\mu}$ are determined by minimizing the expression for the energy

$$E = \int \psi^* H \psi d\tau / \int \psi^* \psi d\tau. \quad (2)$$

This leads to the set of n equations (the secular equations)

$$\sum_{\nu=1}^n (H_{\mu\nu} - ES_{\mu\nu}) x_\nu = 0 \quad (\mu = 1, 2, \dots, n), \quad (3)$$

and thus to the secular determinant

$$\det | H_{\mu\nu} - ES_{\mu\nu} | = 0. \quad (4)$$

Solution of this determinant provides us with the energies E_j ($j = 1, 2, \dots, n$) of the possible orbitals, and (3) gives the corresponding coefficients $x_{j\mu}$. If the m.o. ψ_j is normalized, so that

$$\int \psi_j^2 d\tau = 1, \quad (5)$$

then

$$\sum_{\mu, \nu=1}^n S_{\mu\nu} x_{j\mu} x_{j\nu} = 1. \quad (6)$$

It will be noticed that we have used $x_{j\mu}$ for the coefficients in the m.o. (1). This allows us to reserve the previously used $c_{j\mu}$ for the earlier approximations, in which $S_{\mu\nu} = 0$. As we shall see, there are important relations between $x_{j\mu}$ and $c_{j\mu}$. For the same reason let us use E , or E_j , for the energy found by solution of the full secular determinant (4), and ϵ , or ϵ_j , for the corresponding energy as calculated with neglect of $S_{\mu\nu}$.

NEW DEFINITIONS OF BOND ORDER AND CHARGE DENSITY

We must now proceed to new definitions of bond order and charge density. The old definitions, according to which the mobile bond order $p_{\mu\nu}$ between adjacent atoms μ, ν and the total π -charge q_μ on atom μ , were given by

$$p_{\mu\nu} = \sum_{j=1}^m n_j c_{j\mu} c_{j\nu}, \quad (7)$$

$$q_\mu = \sum_{j=1}^m n_j c_{j\mu}^2, \quad (8)$$

the summation being taken over all the allowed m.o., and n_j being the number of electrons in the j th orbital, will no longer serve. In the first place, the normalization condition (6) shows that $\sum_\mu q_\mu$ would no longer be equal to n , so that if we retained (8)

with $x_{j\mu}$ replacing $c_{j\mu}$, we should be leaving out fractional parts of each electron; and in the second place, for the same reason, p and q would no longer satisfy the same simple differential relations (I, equations (24) and (25))

$$p_{\mu\nu} = \frac{1}{2} \frac{\partial \mathcal{E}}{\partial \beta_{\mu\nu}}, \quad q_\mu = \frac{\partial \mathcal{E}}{\partial \alpha_\mu}.$$

The existence of these relations was quite fundamental to the whole subsequent development of the various self and mutual polarizabilities.

To obtain new definitions we must return to (6), which shows that $S_{\mu\nu}$ may be regarded as the components of the metric tensor of a space defined by the x_μ . This suggests that we should introduce a new set of coefficients $y_{j\mu}$ associated with each ψ_j (or $x_{j\mu}$), and defined by the relations

$$y_{j\mu} = \sum_{\nu=1}^n S_{\mu\nu} x_{j\nu} \quad \text{or} \quad x_{j\mu} = \sum_{\nu=1}^n S^{\mu\nu} y_{j\nu}. \quad (9)$$

We have taken $S^{\mu\nu}$ to be the components of the tensor reciprocal to $S_{\mu\nu}$, so that

$$\left. \begin{aligned} \sum_{\nu=1}^n S^{\lambda\nu} S_{\mu\nu} &= \delta_\mu^\lambda = 1 \quad (\lambda = \mu), \\ &= 0 \quad (\lambda \neq \mu). \end{aligned} \right\} \quad (10)$$

On account of (9) we may call the $y_{j\mu}$ the 'covariant' coefficients associated with ψ_j ; evidently $x_{j\mu}$ are the 'contravariant' coefficients. The introduction of the $y_{j\mu}$ coefficients allows us to write the normalization equation (6) in the more suggestive form

$$\sum_{\nu=1}^n x_{j\nu} y_{j\nu} = 1 \quad (j = 1, 2, \dots, n). \quad (11)$$

In the same way, since the ψ_j correspond to principal directions of the Hamiltonian matrix $H_{\mu\nu}$ ($H_{\mu\nu} = \int \phi_\mu H \phi_\nu d\tau$), the different ψ_j are mutually orthogonal, so that, as may easily be shown directly from the secular equations

$$\sum_{\nu=1}^n x_{i\nu} y_{j\nu} = 0 \quad (i \neq j). \quad (12)$$

We now postulate that the former definitions (7) and (8) of bond order and charge are to be replaced by

$$q_\mu = \sum_{j=1}^m n_j x_{j\mu} y_{j\mu} \quad (\mu = 1, 2, \dots, n), \quad (13)$$

$$p_{\mu\nu} = \frac{1}{2} \sum_{j=1}^m n_j (x_{j\mu} y_{j\nu} + x_{j\nu} y_{j\mu}). \quad (14)$$

In (13) and (14) the summation covers all the occupied m.o. ($j = 1, \dots, m$) and n_j is the number of electrons in the j th orbital.

It is easily verified that when the non-diagonal $S_{\mu\nu}$ are put equal to zero, as in earlier treatments, (13) and (14) reduce to (7) and (8). For, as (9) shows, the distinction between $x_{j\mu}$ and $y_{j\mu}$ disappears, and both reduce to the $c_{j\mu}$ involved in (7) and (8). These $c_{j\mu}$ satisfy the equations (cf. (3) and (6))

$$\sum_{\nu=1}^n (H_{\mu\nu} - \epsilon_j \delta_{\mu\nu}) c_{j\nu} = 0, \quad (15)$$

$$\sum_{\nu=1}^n c_{j\nu}^2 = 1. \quad (16)$$

Equations (15) and (16) show that the new definitions of q_μ and $p_{\mu\nu}$ are indeed generalizations of the old. We shall show later, in (32) to (34) and (39) to (41), that they are satisfactory generalizations because they are successful in preserving the fundamental differential relationships between the total energy and the matrix elements $H_{\mu\nu}$.

CONDITIONS FOR EQUIVALENCE OF OLD AND NEW BOND ORDERS AND CHARGES

Calculations in several special cases, e.g. benzene, butadiene, show that when the overlap integral $S_{\mu\nu}$ is given a constant value S for all nearest neighbours, then the contributions to the bond orders and charge densities from a given m.o. have a value independent of S , and hence the same as in the earlier forms of the theory in which S was completely neglected. This is a special case of a more general result which may be stated in the form that when the matrices $H_{\mu\nu}$ and $S_{\mu\nu}$ commute, the bond orders and charges computed according to (13) and (14) with inclusion of overlap are the same as those computed according to (7) and (8) with neglect of overlap.

First let us show that this condition is necessary. If we suppose that (13) and (14) give the same numerical values as (7) and (8), then

$$x_{j\mu} y_{j\mu} = c_{j\mu}^2, \quad x_{j\mu} y_{j\nu} + x_{j\nu} y_{j\mu} = 2c_{j\mu} c_{j\nu}.$$

From the first of these equations it follows that we may write

$$x_{j\mu}/c_{j\mu} = c_{j\mu}/y_{j\mu} = \lambda_{j\mu}, \quad (17)$$

and now, from the second equation,

$$\lambda_{j\mu}/\lambda_{j\nu} + \lambda_{j\nu}/\lambda_{j\mu} = 2.$$

Hence

$$\lambda_{j\mu} = \lambda_{j\nu} \quad (= \lambda_j, \text{ say}). \quad (18)$$

Thus

$$x_{j\mu} = \lambda_j c_{j\mu}, \quad y_{j\mu} = c_{j\mu}/\lambda_j \quad \text{and} \quad x_{j\mu} = \lambda_j^2 y_{j\mu}. \quad (19)$$

But from the definition of the $y_{j\mu}$ in (9) we have

$$\sum_{\nu=1}^n S_{\mu\nu} \lambda_j c_{j\nu} = c_{j\mu}/\lambda_j,$$

i.e.

$$\sum_{\nu=1}^n \left(S_{\mu\nu} - \frac{1}{\lambda_j^2} \delta_{\mu\nu} \right) c_{j\nu} = 0. \quad (20)$$

This shows that the $c_{j\mu}$ give the principal directions of the tensor $S_{\mu\nu}$. Equation (15) shows that this is also true for $H_{\mu\nu}$. It follows that **S** and **H** commute, as stated to be necessary in the theorem.

This analysis also provides a simple relation between the energy ϵ_j calculated without overlap, and E_j calculated with overlap. For, by (3),

$$\sum_{\nu} H_{\mu\nu} x_{j\nu} = \sum_{\nu} E_j S_{\mu\nu} x_{j\nu},$$

and so, using (9) and (19),

$$\lambda_j \sum_{\nu} H_{\mu\nu} c_{j\nu} = (E_j/\lambda_j) c_{j\mu}.$$

Comparison with (15) shows that

$$\epsilon_j = E_j/\lambda_j^2. \quad (21)$$

We have shown above that a necessary condition that the contributions to bond orders and charge densities from each separate orbital should be the same in the two treatments is that $H_{\mu\nu}$ and $S_{\mu\nu}$ have the same principal directions and hence should commute. It may be shown as follows that the condition is sufficient.

For, if the two tensors have the same principal directions $c_{j\mu}$, then

$$\sum_{\nu} H_{\mu\nu} c_{j\nu} = \epsilon_j c_{j\mu}, \quad \sum_{\nu} S_{\mu\nu} c_{j\nu} = (1/\lambda_j^2) c_{j\mu}.$$

Hence, if E_j is chosen to be equal to $\lambda_j^2 \epsilon_j$, it follows that

$$\sum_{\nu} (H_{\mu\nu} - E_j S_{\mu\nu}) c_{j\nu} = (\epsilon_j - E_j/\lambda_j^2) c_{j\mu} = 0.$$

Thus E_j is the energy when overlap is included, so that the relation (21) is reproduced: and also the $x_{j\mu}$ and $c_{j\mu}$ must be proportional. If we write

$$x_{j\mu} = k c_{j\mu},$$

then the normalizing condition (6) gives that

$$\begin{aligned} 1 &= \sum_{\mu, \nu} S_{\mu\nu} x_{j\mu} x_{j\nu} = k^2 \sum_{\mu, \nu} S_{\mu\nu} c_{j\mu} c_{j\nu} \\ &= k^2 \sum_{\mu} c_{j\mu}^2 / \lambda_j^2 \\ &= k^2 / \lambda_j^2, \end{aligned}$$

if we suppose that the $c_{j\mu}$ are normalized according to (16). From this it follows that $k = \lambda_j$, which gives (19) and thus equal contributions to charges and bond orders in both old and new theories. This establishes both the necessary and sufficient character of the condition stated at the beginning of this section.

SOME PARTICULAR CASES

Before proceeding further, it will be convenient to discuss some particular cases of the applicability or otherwise of the preceding analysis.

(i) If all non-diagonal $S_{\mu\nu}$ are zero, as in the usual development of the theory, then $\mathbf{S}$ is a unit matrix, and quite evidently $\mathbf{S}$ and $\mathbf{H}$ commute, whatever the form of $H_{\mu\nu}$.

(ii) In hydrocarbon molecules it is usual to suppose that all the diagonal terms $H_{\mu\mu}$ are equal, with a value $\alpha (= H_{CC})$, and that all $H_{\mu\nu}$ are zero except when μ and ν are neighbouring atoms, in which case $H_{\mu\nu} = \beta$, for all μ, ν . If we make the corresponding assumption that all $S_{\mu\nu}$ are zero except between neighbours and that then $S_{\mu\nu} = S$ for all μ, ν , it is easily verified that $\mathbf{H}$ and $\mathbf{S}$ commute. This shows that the bond orders calculated for these molecules with neglect of S are entirely unaffected, though as Wheland (1941) has shown, this is not true for the energies.

This particular case is probably the most important of the applications of our theory, and is worth pursuing further. Thus a particular equation of the set (3) is

$$\beta x_{j\pi} + \alpha x_{j\rho} + \beta x_{j\sigma} + \beta x_{j\tau} = E_j y_{j\rho},$$

where π, σ, τ are the neighbouring atoms to ρ . The corresponding equation (9) relating $x_{j\mu}$ and $y_{j\mu}$ is

$$Sx_{j\pi} + x_{j\rho} + Sx_{j\sigma} + Sx_{j\tau} = y_{j\rho}.$$

If we eliminate $x_{j\pi}$ between these two equations, we have

$$(\beta - \alpha S)x_{j\rho} = (\beta - E_j S)y_{j\rho},$$

so that, as in the third equation of (19),

$$x_{j\rho} = \lambda_j^2 y_{j\rho}, \quad (22)$$

$$\text{where} \quad \lambda_j^2 = \frac{\beta - E_j S}{\beta - \alpha S}. \quad (23)$$

A comparison of (21) and (23) now shows that

$$E_j = \frac{\beta \epsilon_j}{\beta + S(\epsilon_j - \alpha)}. \quad (24)$$

Now when $S = 0$, it is well known that the energies ϵ_j are always of the form

$$\epsilon_j = \alpha + m_j \beta, \quad (25)$$

where m_j is some number between ± 3 . It follows that

$$E_j = \frac{\alpha + m_j \beta}{1 + m_j S} = \alpha + \frac{m_j(\beta - \alpha S)}{1 + m_j S}, \quad (26)$$

$$\text{and} \quad \lambda_j = (1 + m_j S)^{-\frac{1}{2}}. \quad (27)$$

Equation (26) was obtained quite differently by Wheland (1941).

(iii) An immediate extension of this latter analysis is to suppose that additional values of $S_{\mu\nu}$ are to be included, so that we consider overlap between atoms other than nearest neighbours. If we suppose that the corresponding $H_{\mu\nu}$ are also included, and that the ratio $H_{\mu\nu} : S_{\mu\nu}$ is the same for all μ, ν (except, of course, when $\mu = \nu$, for which $H_{\mu\mu} = \alpha$, $S_{\mu\mu} = 1$), then $\mathbf{H}$ and $\mathbf{S}$ still commute, for it is soon recognized that if S and β are the numerical values for nearest neighbours,

$$S\mathbf{H} = \beta\mathbf{S} + (S\alpha - \beta)\mathbf{1},$$

where $\mathbf{1}$ is the unit matrix. Charge distributions and bond orders are once again independent of the value of S ; and equations (22) to (27) apply without alteration. Now Mulliken (1948) has given grounds for believing that the ratio $H_{\mu\nu} : S_{\mu\nu}$ is approximately constant for all pairs of atoms and all internuclear distances. The importance of such a fact for our theory is at once apparent.

(iv) The conclusions in (ii) and (iii) above are rendered invalid as soon as any Coulomb term for one atom differs from that for the others. This means that when we are making calculations for heteromolecules, we shall not expect to get valid results unless we include the overlap integral right from the start. Numerical calculation in one or two simple cases suggests that the resulting changes in bond orders are sometimes not very great, so that, for example, the molecular diagrams of pyridine, acridine, carbazole, etc., displayed by Longuet-Higgins & Coulson (1947) though not strictly correct, are unlikely to be much in error.

INTEGRAL FORMULAE FOR CHARGE AND BOND ORDERS

In paper I of this series (equations (42) and (43)) integral formulae were obtained for bond orders and charge densities. These formulae were especially useful in the subsequent discussion of mutual and self-polarizabilities. It becomes important to know whether any such formulae are available in the more general theory here being discussed. We shall now show that these do exist, though in order to obtain them, we are obliged to introduce 'mixed' components of the tensor $H_{\mu\nu}$, defined by

$$H_{\nu}^{\mu} = \sum_{\rho=1}^n S^{\mu\rho} H_{\nu\rho}. \quad (28)$$

These new quantities are not symmetrical in μ and ν , as the $H_{\mu\nu}$ were. But if $\mathbf{H}$ and $\mathbf{S}$ commute, then H_{μ}^{ν} does equal H_{ν}^{μ} . In the case where overlap integrals are neglected, $H_{\mu}^{\nu} = H_{\mu\nu}$.

In terms of these mixed components, the secular determinant (4) may be written

$$\det | H_{\mu}^{\nu} - E\delta_{\mu}^{\nu} | = 0, \quad (29)$$

in which the energy E appears only along the principal diagonal. This determinant is not in general symmetrical about its leading diagonal. Similarly, the secular equations (3) may be written in either of the forms

$$\sum_{\nu=1}^n (H_{\nu}^{\mu} - E\delta_{\nu}^{\mu}) x_{\nu} = 0, \quad \sum_{\mu=1}^n (H_{\mu}^{\nu} - E\delta_{\mu}^{\nu}) y_{\mu} = 0. \quad (30)$$

Equations (29) and (30) are similar to those used in I, so that our subsequent analysis follows formally along exactly the same lines. Thus we write

$$E = \sum_{\mu, \nu} (y_{\mu} H_{\nu}^{\mu} x_{\nu}) / \sum_{\mu} (x_{\mu} y_{\mu}). \quad (31)$$

If a small variation is now made in the H_{ν}^{μ} (treated as all independent), and in the x_{μ} , the variation of E is

$$\delta E = \sum_{\mu, \nu} (y_{\mu} \delta H_{\nu}^{\mu} x_{\nu}) / \sum_{\mu} (x_{\mu} y_{\mu}) + \sum_{\rho} \frac{\partial E}{\partial x_{\rho}} \delta x_{\rho}.$$

Any variations in y_{μ} due to variations in x_{μ} are included in the second term on the right-hand side. The stationary condition imposed on E to give the secular equations (30) shows that $\partial E / \partial x_{\rho} = 0$, so that

$$\left(\frac{\partial E}{\partial H_{\nu}^{\mu}} \right)_{\Delta=0} = x_{\nu} y_{\mu}. \quad (32)$$

Summation over the occupied orbitals gives

$$q_{\mu} = \sum_{j=1}^m n_j \left(\frac{\partial E_j}{\partial H_{\mu}^{\mu}} \right)_{\Delta=0} = \left(\frac{\partial \mathcal{E}}{\partial H_{\mu}^{\mu}} \right)_{\Delta=0}, \quad (33)$$

$$p_{\mu\nu} = \frac{1}{2} \sum_{j=1}^m n_j \left[\frac{\partial E_j}{\partial H_{\mu}^{\nu}} + \frac{\partial E_j}{\partial H_{\nu}^{\mu}} \right]_{\Delta=0} = \frac{1}{2} \left(\frac{\partial \mathcal{E}}{\partial H_{\mu}^{\nu}} + \frac{\partial \mathcal{E}}{\partial H_{\nu}^{\mu}} \right)_{\Delta=0}. \quad (34)$$

We have written $\mathcal{E}$ for the total energy of the mobile electrons, defined by

$$\mathcal{E} = \sum_{j=1}^m n_j E_j, \quad (35)$$

and the notation $()_{\Delta=0}$ means, as in I, that the differentiations involved must always be made under the condition that the various variables satisfy the secular determinant $\Delta = 0$ in (29).

The expressions (33) and (34) may be converted into contour integrals as in I, pp. 46–49. The result is

$$q_{\mu} = 1 - \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\partial}{\partial H_{\mu}^{\mu}} \log \Delta(iy) dy, \quad (36)$$

$$\begin{aligned} p_{\mu\nu} &= (-1)^{\mu+\nu+1} \frac{1}{2\pi} \int \frac{\Delta_{\mu, \nu}(iy) + \Delta_{\nu, \mu}(iy)}{\Delta(iy)} dy \\ &= -\frac{1}{2\pi} \int_{-\infty}^{\infty} \left(\frac{\partial}{\partial H_{\mu}^{\nu}} + \frac{\partial}{\partial H_{\nu}^{\mu}} \right) \log \Delta(iy) dy. \end{aligned} \quad (37)$$

There is a similar integral relation for $\mathcal{E}$. Thus (cf. I (44))

$$\mathcal{E} = \sum H_{\mu}^{\mu} + \frac{1}{\pi} \int_{-\infty}^{\infty} \left\{ iy \frac{\Delta'(iy)}{\Delta(iy)} - n \right\} dy. \quad (38)$$

In these expressions $\Delta(iy)$ stands for the unsymmetrical determinant of (29) with iy replacing E ; $\Delta_{\mu, \nu}(iy)$ is the same determinant with row μ and column ν struck out. In (36) to (38) we have tacitly assumed that the occupied orbitals are all bonding, and the unoccupied are anti-bonding, so that $E = 0$ lies between E_m and E_{m+1} . This is nearly always the case, but may not always be so if some of the heteroatoms present have

very large Coulomb terms. If this should happen, the determinants in these equations should be regarded as functions of $x + iy$, rather than iy , where x is any value between E_m and E_{m+1} . Provided that x lies in this range, all three integrals are actually independent of its precise value. Formulae (36) to (38) are similar to (42) to (44) of paper I, and do indeed reduce to them when overlap integrals are neglected. The most significant difference is that our new formulae involve the components H_{μ}^{ν} of the mixed tensor rather than the components $H_{\mu\nu}$ of the original tensor. This suggests very strongly that it is these mixed components H_{μ}^{μ} and H_{μ}^{ν} that have physical significance rather than the simple Coulomb integrals $H_{\mu\mu}$ and resonance integrals $H_{\mu\nu}$. Now the essential difference between H_{μ}^{μ} and $H_{\mu\mu}$ is that whereas the latter is a function only of atom μ , the former depends not only on μ but also on all the neighbours of μ . In this respect, then, our new interpretation is likely to be better than the old one, for as was explained in I, p. 43, we expect the diagonal elements $H_{\mu\mu}$ (or H_{μ}^{μ}) to be a measure of the electron affinity of atom μ ; and it is at least plausible that this electron affinity should depend not only on μ , but also, to a lesser extent, on the neighbours of μ . Some recent calculations by Mulliken (1948*b*) and Löwdin (1948) reinforce this belief. This argument is concerned, of course, solely with what happens in non-polar molecules, as a result of the geometrical arrangement of atoms around any given atom. It is well recognized that in polar molecules there are other effects which become important, and to which we have not referred here. The most significant of these is the fact that as a result of the charge displacements in such molecules, the Coulomb terms α depend on the final charge distribution, and may not be taken as constants in the way that we have done.

There is another reason for preferring our new correlation with quantities of physical significance. This may be seen most conveniently if we write down the values of H_{μ}^{ν} for the two particular molecules butadiene and benzene. The numerical values shown below are calculated on the assumptions that only nearest-neighbour interactions need be considered, and that all $H_{\mu\mu} = \alpha$, $H_{\mu\nu} = \beta$, $S_{\mu\mu} = 1$, $S_{\mu\nu} = 0.25$ (Mulliken, Rieke & Brown 1941). In these expressions γ stands for the combination

$$\gamma = \beta - \alpha S. \quad (39)$$

butadiene:

$$S_{\mu\nu} = \begin{pmatrix} 1 & S & . & . \\ S & 1 & S & . \\ . & S & 1 & S \\ . & . & S & 1 \end{pmatrix}, \quad H_{\mu\nu} = \begin{pmatrix} \alpha & \beta & . & . \\ \beta & \alpha & \beta & . \\ . & \beta & \alpha & \beta \\ . & . & \beta & \alpha \end{pmatrix}$$

$$H_{\mu}^{\nu} = \begin{pmatrix} \alpha - 0.287\gamma & 1.148\gamma & -0.305\gamma & 0.076\gamma \\ 1.148\gamma & \alpha - 0.594\gamma & 1.224\gamma & -0.305\gamma \\ -0.305\gamma & 1.224\gamma & \alpha - 0.594\gamma & 1.148\gamma \\ 0.076\gamma & -0.305\gamma & 1.148\gamma & \alpha - 0.287\gamma \end{pmatrix}.$$

benzene:

$$S_{\mu\nu} = \begin{pmatrix} 1 & S & . & . & . & S \\ S & 1 & S & . & . & . \\ . & S & 1 & S & . & . \\ . & . & S & 1 & S & . \\ . & . & . & S & 1 & S \\ S & . & . & . & S & 1 \end{pmatrix},$$

$$H_{\mu\nu} = \begin{pmatrix} \alpha & \beta & . & . & . & \beta \\ \beta & \alpha & \beta & . & . & . \\ . & \beta & \alpha & \beta & . & . \\ . & . & \beta & \alpha & \beta & . \\ . & . & . & \beta & \alpha & \beta \\ \beta & . & . & . & \beta & \alpha \end{pmatrix},$$

$$H_{\mu}^{\nu} = (\alpha - 0.622\gamma, 1.241\gamma, -0.356\gamma, 0.178\gamma, -0.356\gamma, 1.241\gamma).$$

Only the top rows of the matrices for benzene need to be given; subsequent ones are obtained by simple permutation.

One interesting thing about these H_μ^ν is that the off-diagonal elements are all simple multiples of $\gamma = \beta - \alpha S$. This is a point in their favour, for Mulliken (1948*a*) has recently stressed the importance of this expression, originally introduced by Mulliken *et al.* (1941). Another pleasing thing about these figures is the variation shown in the leading diagonal elements, which are all of the form $\alpha - k\gamma$, where k has the values 0.287, 0.594 for C_1 and C_2 in butadiene, and 0.622 for all the carbons in benzene. The fact that the diagonal element is numerically larger at the end atoms than at the central ones in butadiene is in good accord with the calculations of Mulliken (1948*b*), who showed, from a comparison of graphite, benzene and ethylene, where each carbon atom has 3, 2 or 1 other carbon neighbours, that the more surrounded an atom was with similar neighbours, the smaller would be the numerical value of its diagonal element in the secular determinant.

BOND AND CHARGE AFFINITIES

We have just demonstrated the relevance of the components H_μ^ν of the mixed tensor. It is desirable that we should have some name by which we may refer to them, and distinguish them from the $H_{\mu\nu}$. We suggest that H_μ^μ should be called the *charge affinity* of atom μ , and H_μ^ν the *bond affinity* of μ towards ν . Since $H_\mu^\nu \neq H_\nu^\mu$, we have to be careful in the use of these bond affinities. Our convention is that the lower suffix indicates the atom from which the bond is regarded as starting, and that the upper suffix indicates the direction towards which the bond $\mu - \nu$ is directed. Our reasons for the choice of these names will appear shortly.

Just as the one resonance integral $H_{\mu\nu}$ has had to be replaced by two bond affinities H_μ^ν and H_ν^μ , so we may split up the bond order $p_{\mu\nu}$ into two parts p_μ^ν and p_ν^μ associated with the two ends μ and ν of the bond. The definitions of these quantities (see (33) and (34)) are

$$p_\mu^\nu = \frac{\partial \mathcal{E}}{\partial H_\mu^\nu}, \quad p_\nu^\mu = \frac{\partial \mathcal{E}}{\partial H_\nu^\mu}, \quad q_\mu = \frac{\partial \mathcal{E}}{\partial H_\mu^\mu}. \quad (40)$$

The bond order $p_{\mu\nu}$ is the average of the two parts which we have just introduced, so that

$$p_{\mu\nu} = \frac{1}{2}(p_\mu^\nu + p_\nu^\mu). \quad (41)$$

With these definitions, the total energy of the molecule may be written

$$\mathcal{E} = \sum_{\mu=1}^n q_\mu H_\mu^\mu + \sum_{\mu, \nu=1}^n p_\mu^\nu H_\mu^\nu. \quad (42)$$

This last equation could have been arrived at in a different way. For, by (31) and (11), if the m.o. is normalized:

$$E_j = \sum_{\mu, \nu} x_{j\mu} y_{j\nu} H_\nu^\mu.$$

Thus

$$\begin{aligned} \mathcal{E} &= \sum_{\mu, \nu} n_j x_{j\mu} y_{j\nu} H_\nu^\mu \\ &= \sum_{\mu} q_\mu H_\mu^\mu + \sum_{\mu, \nu} p_\mu^\nu H_\mu^\nu, \quad \text{as required.} \end{aligned}$$

Equation (42) provides the justification for our choice of the names charge- and bond-affinity. Thus, if we have a single charge on atom μ this contributes H_μ^μ to the

energy. This is obviously closely related to what is usually called the electron affinity of atom μ . We have therefore called it the charge affinity of atom μ . Since H_μ^μ —unlike the conventional electron affinity—depends on the other atoms of the molecule as well as on atom μ , it would be more strictly accurate to refer to it as the charge affinity of μ in its position in the given molecule. Exactly similar considerations justify calling H_μ^ν the bond affinity of μ towards ν .

It will be noticed from the table of H_μ^ν for butadiene and benzene given earlier, that bond affinities exist even between atoms so far apart that we can neglect their overlap $S_{\mu\nu}$. Examples are the *para* positions in benzene and the 1, 4 positions in butadiene. It is certainly in accord with chemical intuition that there should be an interaction of this kind and that it should contribute to the energy of the molecule. Nor is it surprising that some of these H_μ^ν are positive and others negative; and that in a conjugated polyene chain their magnitudes decrease as μ and ν are more distant from one another.

Since, from (28), H_μ^ν involves $S^{\mu\nu}$, and (10) shows that $S^{\mu\nu}$ is the inverse matrix of $S_{\mu\nu}$, there is in general no simple expression for H_μ^ν . But in the case of a polyene chain of $2n$ similar atoms numbered consecutively from 1 to $2n$, it may be shown that, neglecting all overlap except between nearest neighbours,

$$H_\mu^\nu = (-1)^{\mu+\nu+1} 4\gamma \cosh^2 \theta \sinh \mu \theta \sinh (2n - \nu + 1) \theta / \sinh \theta \sinh (2n + 1) \theta, \quad (43)$$

where $\cosh \theta = 2$ and it is supposed that $\mu < \nu$. The case of $\mu > \nu$ is obtained at once from the fact that in this problem H_μ^ν is symmetrical in μ and ν . The formula (43) illustrates the alternating character of the H_μ^ν . It also shows that H_μ^ν decreases if ν increases or if μ decreases, corresponding to an increased separation of μ and ν along the chain. It may be added that (43) applies equally, with n instead of $2n$, for even or odd values of n .

POLARIZABILITIES

The bond and charge affinities just discussed enable us to generalize the various self- and mutual-polarizabilities which were introduced in I, and which measure the changes produced in charges or bond orders when any of these affinities is altered.

(i) The atom-atom polarizability gives the change in charge density q_μ induced by a change in the charge affinity H_ν^ν . Thus (cf. I, equation (46))

$$\pi_{\mu,\nu} = \frac{\partial q_\mu}{\partial H_\nu^\nu} = \frac{\partial^2 \mathcal{E}}{\partial H_\mu^\mu \partial H_\nu^\nu} = \pi_{\nu,\mu}. \quad (44)$$

(ii) The bond-atom polarizability gives the change in bond order $p_{\rho\sigma}$ induced by a change in charge affinity H_μ^μ . Thus

$$\pi_{\rho\sigma,\mu} = \frac{\partial p_{\rho\sigma}}{\partial H_\mu^\mu}. \quad (45)$$

(iii) The atom-bond polarizability is

$$\pi_{\mu,\rho\sigma} = \frac{\partial q_\mu}{\partial H_\rho^\sigma} + \frac{\partial q_\mu}{\partial H_\sigma^\rho} = \frac{\partial^2 \mathcal{E}}{\partial H_\rho^\sigma \partial H_\mu^\mu} + \frac{\partial^2 \mathcal{E}}{\partial H_\sigma^\rho \partial H_\mu^\mu} = 2\pi_{\rho\sigma,\mu}. \quad (46)$$

(iv) Finally the bond-bond polarizability is

$$\pi_{\mu\nu,\rho\sigma} = \frac{\partial p_{\mu\nu}}{\partial H_\rho^\sigma} + \frac{\partial p_{\mu\nu}}{\partial H_\sigma^\rho} = \pi_{\rho\sigma,\mu\nu}. \quad (47)$$

The formal resemblance between (44) to (47) and the corresponding equations of I shows that integral expressions completely analogous to I, (52) to (54), can be written down for these quantities, in the manner of (36) and (37). We do not need to reproduce these here.

There is one point concerning equations (44) to (47) which needs further discussion. We have assumed, in making the differentiations, that all the H_μ^ν are independent variables. This assumption was also involved in (32) to (34) and (35) to (38). At first sight this seems an incorrect procedure because the H_μ^ν are derived (see (28)) from the $H_{\mu\nu}$, and there are only $n(n+1)/2$ independent quantities of this kind, on account of the symmetry relation $H_{\mu\nu} = H_{\nu\mu}$. Yet we have treated all n^2 of the H_μ^ν as independent. The justification of our procedure lies in the fact that a complete description of the molecule requires both $H_{\mu\nu}$ and $S_{\mu\nu}$. Now there are $n(n-1)/2$ coefficients $S_{\mu\nu}$ which may be changed by perturbations in the molecule. This is because $S_{\mu\mu} = 1$, and $S_{\mu\nu} = S_{\nu\mu}$. Thus the total number of independent parameters is

$$n(n+1)/2 + n(n-1)/2 = n^2,$$

and we may equally well choose these to be either (a) $H_{\mu\nu}$ and $S_{\mu\nu}$, or (b) H_μ^ν . We have chosen to regard the H_μ^ν as of physical significance rather than the $H_{\mu\nu}$, so the alternative (b) is naturally the one to adopt. We can thus treat all the H_μ^ν as independent parameters, and must realize that the small changes, or perturbations, involved in (40) to (42) could, if we wished, be regarded as small changes both in the $H_{\mu\nu}$ and in the $S_{\mu\nu}$.

There are, of course, alternative expressions for the polarizabilities (44) to (47) in terms of the coefficients $x_{j\mu}$ and $y_{j\mu}$ of the allowed molecular orbitals. These expressions, which are often more convenient to use in a particular example than are the integral representations, may be found by first-order perturbation theory in a manner somewhat similar to that of I, p. 53. On account of the non-orthogonality of the ϕ_μ , the analysis is a little more complicated.

We start with the original secular equations (30) involving the H_ν^μ ; and we suppose that each H_ν^μ receives an increment $H_\nu^{\prime\mu}$, the corresponding increments in $x_{j\nu}$ and $y_{k\mu}$ being $x'_{j\nu}$ and $y'_{k\mu}$. All the 'dashed' quantities are small. Then by straightforward, though somewhat tedious, algebra in which we consider merely terms of the first order of small quantities, it may be shown that

$$x'_{j\mu} = \sum_{k,\nu,\alpha} \frac{x_{j\nu} y_{k\alpha} x_{k\mu} H_\nu^{\prime\alpha}}{E_j - E_k}, \quad (48)$$

and

$$y'_{j\mu} = \sum_{k,\nu,\alpha} \frac{x_{k\nu} y_{j\alpha} y_{k\mu} H_\nu^{\prime\alpha}}{E_j - E_k}. \quad (49)$$

In the course of the proof, which we shall not reproduce in full, we require to use the facts that

$$\sum_\nu x_{j\nu} y_{k\nu} = \sum_\nu x_{k\nu} y_{j\nu}, \quad (50)$$

$$\sum_{\nu,\mu} x_{j\nu} H_\nu^\mu y_{k\mu} = \sum_{\nu,\mu} x_{k\nu} H_\nu^\mu y_{j\mu}, \quad (51)$$

$$\sum_k x_{k\mu} y_{k\nu} = \delta_\mu^\nu. \quad (52)$$

The first two of these may be proved from the transformation properties (9) and (28) for the $x_{j\mu}$ and $y_{j\mu}$. The third is a consequence of geometrical relationships in the space defined by the x_μ , such as those involved in proving equation (12); it may, however, be proved without difficulty by proper use of the orthogonality and normalization equations (11) and (12).

Once we have calculated the changes in the quantities $x_{j\mu}$, $y_{j\mu}$ due to the perturbation, it is easy to determine the changes in any q_μ or $p_{\mu\nu}$, using the definitions of these quantities given in (13) and (14). The method is exactly analogous to I, pp. 53–55, and the results are:

$$\pi_{\mu,\nu} = 2 \sum_{j=1}^m \sum_{k=1}^n \frac{x_{j\nu}x_{k\mu}y_{j\mu}y_{k\nu} + x_{j\mu}x_{k\nu}y_{j\nu}y_{k\mu}}{E_j - E_k} = \pi_{\nu,\mu} \quad (53)$$

$$\pi_{\mu,\rho\sigma} = 2 \sum_{j=1}^m \sum_{k=1}^n \frac{x_{k\mu}y_{j\mu}(x_{j\rho}y_{k\sigma} + x_{j\sigma}y_{k\rho}) + x_{j\mu}y_{k\mu}(x_{k\rho}y_{j\sigma} + x_{k\sigma}y_{j\rho})}{E_j - E_k} = 2\pi_{\rho\sigma,\mu} \quad (54)$$

$$\pi_{\mu\nu,\rho\sigma} = \sum_{j=1}^m \sum_{k=1}^n \frac{(x_{j\mu}y_{k\nu} + x_{j\nu}y_{k\mu})(x_{k\rho}y_{j\sigma} + x_{k\sigma}y_{j\rho}) + (x_{k\mu}y_{j\nu} + x_{k\nu}y_{j\mu})(x_{j\rho}y_{k\sigma} + x_{j\sigma}y_{k\rho})}{E_j - E_k} = \pi_{\rho\sigma,\mu\nu} \quad (55)$$

Since all these formulae have numerators that are symmetrical in the suffixes j and k , and denominators that are antisymmetrical, the k -summation need only be taken over the unoccupied orbitals, so that k runs from $m+1$ to n .

These formulae all reduce to the corresponding values (64) to (67) already found in I, when $S_{\mu\nu}$ is neglected; for then $x_{j\mu} = y_{j\mu} = c_{j\mu}$. Further, when $\mathbf{S}$ and $\mathbf{H}$ commute, so that bond orders and charges are independent of overlap, the numerators all have the same values as in I, and so the only differences arise from the altered denominators. The particular case of a hydrocarbon with nearest-neighbour interaction is sufficiently important to be worth describing a little more fully. According to (25) and (26)

$$\epsilon_j = \alpha + m_j\beta, \quad E_j = \frac{\alpha + m_j\beta}{1 + m_jS} = \lambda_j^2\epsilon_j,$$

$$\text{so that} \quad \epsilon_j - \epsilon_k = (m_j - m_k)\beta, \quad (56)$$

and, after a little reduction,

$$E_j - E_k = \lambda_j^2\lambda_k^2(m_j - m_k)\gamma \quad [\gamma = \beta - \alpha S]. \quad (57)$$

Hence the formulae for such hydrocarbons are merely modified by the insertion of $\lambda_j^2\lambda_k^2$ in the denominators of each term, the result now being given as a multiple of $1/\gamma$ instead of $1/\beta$ as in the earlier theory.

Tables 1 and 2 show the atom-atom polarizabilities calculated according to (53) for benzene and butadiene, the values of S and H being the same as on p. 204.

TABLE 1. ATOM-ATOM POLARIZABILITIES IN BENZENE

	$\pi_{1,1}$	$\pi_{1,2}$	$\pi_{1,3}$	$\pi_{1,4}$
new value, with overlap, units of $1/\gamma$	0.359	-0.137	0.0081	-0.100
old value, without overlap, units of $1/\beta$	0.398	-0.157	0.0093	-0.102

TABLE 2. ATOM-ATOM POLARIZABILITIES IN BUTADIENE

$\pi_{\mu,\nu}$	new values, including S		old values, without S	
	$\nu=1$	$\nu=2$	$\nu=1$	$\nu=2$
$\mu=1$	0.601	-0.377	0.626	-0.402
$=2$	-0.377	0.377	-0.402	0.402
$=3$	0.042	-0.042	0.045	-0.045
$=4$	-0.266	0.042	-0.268	0.045

It is very interesting to find that, notwithstanding the new definitions of polarizabilities, there is really extremely little difference between the two sets of figures. This is particularly gratifying because it justifies us in making our calculations of polarizabilities using the simpler theory, and believing that qualitative conclusions therefrom will be quite unaltered, and quantitative ones only slightly affected. In view of the more clumsy character of the complete theory, the changes are scarcely great enough to warrant its use, except in special cases where considerable accuracy is required.

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